

Natalie Leaist

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EDUCATION

University of British Columbia

Vancouver, BC, Canada

Bachelor of Applied Science in Engineering Physics, minor in Honours Math

September 2023 - Present

- 94% cumulative average (4.33/4.33 GPA), 96% average across math and physics coursework
- Trek Excellence Scholarship recipient (top 5% of faculty and year), EnSURE Award recipient & Dean's Scholar

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript

Mathematical Methods: Least Squares, SVD, PCA, FFT/DFT, Numerical Integration, Linear Regression

Embedded Systems: Microcontrollers, FreeRTOS, Communication Protocols, Sensors, Arduino

Electrical: Circuit Design, PCB Design, KiCad, Oscilloscope, Logic Analyzer, AWG

Other Tools: Git, Linux, VSCode, OpenCV, L^AT_EX, MATLAB, Simulink

EXPERIENCE

Research Intern & Software Developer

May 2025 – August 2025

Experimental Cosmology Group, UBC Physics & Astronomy

Vancouver, BC

- Worked on the CGEM project, a radio telescope that will be used to help detect gravitational radiation from the early universe (collaboration between UBC, NASA Goddard Space Flight Centre and DRAO)
- Developed a high-frequency (830 Hz) embedded data acquisition system for accelerometer-based vibration and inclination monitoring of the telescope using an STM32 microcontroller, AD7177-2 ADC, DMA, SPI communication, FreeRTOS, and Ethernet streaming
- Performed Python-based FFT, spectral noise analysis, and calibration, to verify accelerometer performance and achieve less than 1 μm displacement resolution
- Optimized embedded timing, memory alignment, and signal integrity, improving system reliability and throughput

Control Systems & Software Developer

September 2024 – Present

UBC Orbit Satellite Design Team

Vancouver, BC

- Working on the ALEASAT project, an Earth observation satellite supported by the European Space Agency's Fly Your Satellite! Program
- Creating a Helmholtz cage to simulate the magnetic field of low Earth orbit
- Developing Python simulations of satellite attitude dynamics and control laws
- Constructed firmware and software to fulfill mission and testing requirements set by the European Space Agency
- Implemented CPU usage monitoring across FreeRTOS tasks

Outdoor Adventure Guide

July 2022 – August 2023

The Adventure Group

Whistler, BC

- Led groups of up to 20 guests on Canada's highest and 2nd longest zip-line
- Received over 50 five-star guest reviews, the highest among guides, highlighting strong communication and leadership
- Conducted safety inspections on off-road vehicles, zip-lines, and aerial obstacle courses
- Trained in leadership, emergency response, risk assessment, and guest management

PROJECTS

Slow-Light Analog in Coupled RLC Circuits

- Investigated how a chain of coupled RLC resonators can mimic slow-light behavior observed in dispersive optical media
- Analyzed how resonance modifies group delay and attenuation using the system transfer function
- Constructed a prototype RLC network to observe qualitative pulse delay behavior

Self-Balancing Robot

- Designed and built a two-wheeled self-balancing robot using Arduino, MPU6050 IMU, DRV8825 motor drivers, and NEMA-17 stepper, gaining hands-on experience in hardware integration and embedded programming
- Implemented a complementary filter to combine accelerometer and gyroscope data, providing a stable pitch estimate and developing practical skills in sensor fusion and noise reduction
- Developed and tuned a PID controller to actively stabilize the robot's pitch

ACTIVITIES

Freeride Skiing

- Ranked 2nd in Canada (2022-2023 season), qualifying for Junior World Championships (2023) as one of 15 junior female skiers worldwide
- Recognized with Rossignol Freeride Platinum Sponsorship (Sept. 2023 – Aug. 2024), the highest athlete tier before Rossignol's professional team